

Experiments 1A–1D: Long-form Laboratory Report

Mathematical Gate, Discovery Atlas, Adaptive Refinement, and Trainability
Query–Key Initialization Geometry Research Program

Written to be shortened later into a research paper

Automated report from frozen artifacts in
`code/exp1a/data/full`, `code/exp1b/data/full`,
`code/exp1c/data/full`, and `code/exp1d/data/full`

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Chapter 1

How to Read This Document

This file is **not** a conference paper. It is a laboratory memoir: longer, slower, and more explicit than the paper that will later be cut from it. The point of writing this way first is honesty. If a later paper is shorter, it should be shorter because we *removed* repetition and moved details to an appendix, not because we quietly strengthened a claim.

1.1 What this document is for

A research paper has a tight page budget. That budget rewards compact sentences, omitted caveats, and figures that look decisive. Those habits are useful *after* the evidence is already understood. They are dangerous *while* the evidence is still being digested.

This report therefore does four jobs the paper will not have room to do:

1. Explain the scientific question in ordinary words before introducing notation.
2. Say what each experiment is *for*, not only what it measured.
3. Keep every important number next to the interpretation it actually supports.
4. Separate four kinds of statement: exact mathematics, a measurement, an interpretation, and something not yet established.

When the paper is written, the right workflow is: copy a claim from here, then shorten the prose. Do not invent a cleaner story than the one below.

1.2 A one-page map of the four experiments

The whole program asks a single question:

When we initialize a Transformer's query and key matrices, their *relationship* to each other is a free choice. Does that choice create distinct attention behaviors at initialization, and do those behaviors predict early training?

The four stages answer four different sub-questions:

- 1A Is the theory coded correctly? If the formulas are wrong in software, every later map is fiction. This stage **passed**.
- 1B If we sweep the allowed initializations, what kinds of attention do we actually see? This produced a discovery atlas: a map, not a proof of trainability.
- 1C If we zoom in on uncertain map cells and also try a wider/longer geometry, do “good-looking” cells stay good-looking? They mostly do not, jointly. A weaker freeze rule was used to pick eight heads for later work.
- 1D If we freeze those labels *before* looking at training loss, do the “screened” cells train better than a pool of “pathology” cells on a tiny one-block decoder? Associative recall: no. TinyStories: yes under the preregistered pooling rule, but the yes is mainly a scale effect, not a geometry effect.

1.3 How to use this report when writing the paper

Keep the following compression rules in mind:

- **Keep the mixed 1D result.** Do not let TinyStories “PASS” become “the atlas predicts trainability.”
- **Keep the s -transfer warning.** s is already a multiplier on $1/\sqrt{d}$, not an absolute variance. It still does not transfer, because copying the same s changes both entry noise ($\sigma = s/\sqrt{d}$) and typical attention scores ($s^2\sqrt{m}$). Making s an absolute variance would not fix that.
- **Keep the 1C fallback.** The original freeze rule found zero surviving cells. The eight-head file exists only because a documented weaker rule was used.
- **Do not upgrade labels.** “Screened candidate” means “looked healthy at initialization under our screens.” It does not mean “will train well” and does not mean “better than diffuse at the same scale.”

Chapter 2

Executive Summary

In plain language. We checked the math, drew a map of initialization behaviors, tried to refine the map, then asked whether the map’s “healthy” region trains better than a mixed “unhealthy” region on a tiny decoder. The math checks out. The map has real structure. The refined map does not survive a strict transfer test. Tiny training helps only in a limited, easy-to-overclaim way.

This document records the confirmatory analysis for the first four stages of the initialization-atlas program.

- **Experiment 1A** (mathematical and implementation gate): **PASSED**. All 864 Monte Carlo moment configurations and all auxiliary integration checks succeeded under preregistered tolerances. In other words: the formulas for attention-score means and variances, and the surrounding identities, match the code.
- **Experiment 1B** (discovery atlas): **COMPLETE**. The full 1,717-cell coefficient grid was measured at reference geometry $(d, m, n) = (64, 16, 64)$ across 879,104 raw forward/backward records; bootstrap analysis produced 13,736 stratified summaries. In other words: we have a dense map of how attention looks at initialization, under several input and mask conditions.
- **Experiment 1C** (adaptive refinement + transfer): **MEASURED AND FROZEN UNDER FALLBACK**. 120 surrogate-chosen cells $\times$ 2 geometries = 240 tasks, 122,880 raw rows, 1,920 stratified summaries. The preregistered all-stratum intersection is empty (0/120). The documented fallback froze 8 heads from 32 majority-screened reference-geometry points. In other words: we asked a cell to look “screened” in every repeated setting at once, and none did. We then used a weaker, pre-written backup rule so that Experiment 1D would have something to train.
- **Experiment 1D** (one-block trainability): **COMPLETE**. 84 points, 1,344 jobs, 0 divergences. Gate **fails** on associative recall and **passes** on TinyStories. The pass is a within-1D larger s (fan-in multiplier), not transferable 1B labels. Weights use $s/\sqrt{d}$; logits scale as $s^2\sqrt{m}$. In other words: on children’s-story next-byte prediction, quieter initializations improved more than much louder ones. On a synthetic key-value memory task, every initialization looked the same because the probe was saturated.

2.1 Headline 1B findings

These counts pool all 13,736 bootstrap-labeled summaries. A “stratum” is one coefficient cell measured under one input/mask condition, not one unique (s, α, ϕ) point.

Label	Count	Fraction
Diffuse	5,362	39.0%
Concentrated	3,793	27.6%
Screened candidate	2,867	20.9%
Boundary / uncertain	1,241	9.0%
Self-locked	472	3.4%
Gradient-starved	1	< 0.01%

What the names mean, in one line each:

- **Diffuse:** attention is almost uniform. Every key gets about the same weight.
- **Concentrated:** attention is peaky. One or a few keys dominate the row.
- **Self-locked:** the token mostly attends to itself.
- **Screened candidate:** not obviously pathological, and entropy is in a middle band.
- **Boundary / uncertain:** our 95% intervals sit on both sides of a threshold, so we refuse to pick a side.
- **Gradient-starved:** a one-step backward probe looks too quiet, relative to a baseline floor.

These labels are **descriptive** statements about initialization-time attention geometry. They are *not* claims about downstream language-model quality until validated in Experiment 1D, and even 1D does not promote them into a 32M-model result.

2.2 Headline 1C findings

$N = 1,920$ adaptive/transfer summaries. The adaptive sampler was *asked* to spend budget on uncertain cells and theory-disagreement cells, so the absence of concentrated and self-locked pathology is by design, not a discovery that those regions vanished.

Label	Count	Fraction
Diffuse	793	41.3%
Boundary / uncertain	613	31.9%
Screened candidate	514	26.8%
Concentrated / self-locked / gradient-starved	0	0%

Transfer majority-label agreement between $(64, 16, 64)$ and $(128, 16, 128)$ is only 30/120 coefficient points. That is a low agreement rate: most of the 120 coefficient choices do not keep the same majority label when width and sequence length change. The 1C freeze used the documented fallback: majority screening on the reference geometry, not the empty all-stratum intersection.

2.3 Headline 1D findings

84 frozen points, 1,344 completed jobs, 10,000 paired bootstrap draws. The primary statistic is:

(mean loss decrease per token among screened cells) minus (the same quantity among pooled pathology cells).

A task passes only if the 95% confidence interval for that difference lies entirely above zero. “Above zero” means screened cells improved faster, per token, than the pathology pool.

Task	$\hat{\Delta}$	95% CI	Gate	Implied total-loss gap
Associative recall	-1.16×10^{-7}	$[-4.68, +2.47] \times 10^{-7}$	FAIL	-0.015 nats
TinyStories	$+2.06 \times 10^{-8}$	$[+1.86, +2.32] \times 10^{-8}$	PASS	+0.338 nats

The TinyStories pass is real under the preregistered pooling rule, but it is **not** evidence that screened cells beat scale-matched diffuse cells, and it is **not** a transferable statement about numerical s . Constructor entries already use $1/\sqrt{d}$; s is only a multiplier on that fan-in Gaussian. 1B pathology fill had median $s = 2.0$ and 2.83 at $(d, m) = (64, 16)$; 1D applied those same numbers at $(128, 32)$. That changes both $\sigma = s/\sqrt{d}$ and logit mean $\propto s^2\sqrt{m}$. On TinyStories, screened and diffuse (same 1C s -band) finish at 1.360 vs. 1.365; concentrated and self-locked finish at 1.619 and 2.095. Associative recall is undifferentiated (all strata ≈ 4.172) and clipped on every step.

Chapter 3

Scientific Context and Motivation

3.1 The everyday picture

A Transformer does not read a sentence the way a human does. At each position, it builds a weighted average of other positions. Those weights are called **attention**. If the weights are almost equal, the model is averaging everyone together (diffuse). If one weight is huge, the model is focusing (concentrated). If the huge weight is usually on the token itself, the model is mostly looking in a mirror (self-locked).

Those weights come from a comparison between two learned views of the same tokens:

- a **query** view: “what am I looking for?”
- a **key** view: “what do I contain?”

The comparison is a scaled dot product. If query and key are initialized independently, their typical comparison is one thing. If they are initialized as almost the same matrix, the comparison is another thing. That difference is the subject of this program.

In plain language. Two models can have equally noisy query weights and equally noisy key weights, and still behave differently, because the *joint* pattern of those two matrices can make attention scores systematically larger on the diagonal, or noisier, or both.

3.2 What problem is being studied?

Transformer attention computes, for each query position i and key position j ,

$$\ell_{ij} = \frac{1}{\sqrt{m}} (W_q x_i)^\top (W_k x_j), \quad P_{ij} = \frac{\exp(\ell_{ij})}{\sum_{j'} \exp(\ell_{ij'})},$$

where $W_q, W_k \in \mathbb{R}^{m \times d}$ are query and key projections and $x_i, x_j \in \mathbb{R}^d$ are token representations.

The first line is the **logit** (attention score before softmax). The second line is the **attention probability**. Softmax turns scores into weights that sum to one.

Standard initializers (for example Xavier or Kaiming on independent Gaussian weights) fix **marginal** variances: they control how large typical entries of W_q are, and separately how large

typical entries of W_k are. They do not explicitly control the **joint geometry** of (W_q, W_k) : whether those two matrices point the same way, point in orthogonal leftover directions, or are unrelated.

The research program introduces a **correlated Gaussian–orthogonal** family parameterized by $(s, \alpha, \beta, \gamma)$ (and an angle ϕ that encodes β and γ). It derives **exact finite-width** expressions for logit moments: formulas that do not require the infinite-width limit. It then **maps** where different attention behaviors occur in coefficient space, and only afterwards tests trainability on a small decoder.

3.3 Why “geometry” is not a metaphor for 3-D space

In this report, “geometry” means the statistical shape of the query-key product

$$M = W_q^\top W_k.$$

Attention logits are $x_i^\top M x_j / \sqrt{m}$. So the object that actually sets the operating point of softmax is M , not W_q or W_k taken separately. Two initializations can match

$$\mathbb{E}\|W_q\|_F^2 = \mathbb{E}\|W_k\|_F^2$$

and still produce different laws for M . That is the degree of freedom being mapped.

3.4 Why staged experiments before training a large language model?

Training a 32M language model is expensive, noisy, and easy to over-interpret. If we jump there first, we will not know whether a failure came from (i) a coding error in the initializer, (ii) an initialization-time attention pathology, (iii) a scale mismatch, or (iv) ordinary optimization and data effects.

The stages are therefore blocking questions, in order:

1. **1A** answers: “Is the theory implemented correctly?” If not, every map is meaningless.
2. **1B** answers: “Where in (s, α, ϕ) do diffuse, concentrated, self-locked, and screened behaviors occur at initialization?” This is an empirical atlas, not a hand-drawn phase diagram.
3. **1C** answers: “Do uncertain/boundary cells remain screened under a fitted surrogate, extra samples, and a width/length transfer?” A frozen portfolio is required before 1D, so that we cannot pick winners after seeing training curves.
4. **1D** answers: “Do screened candidates reduce early training loss relative to pooled pathology groups on a one-block decoder?” A confidence interval excluding zero in the preregistered direction is required to call the atlas useful for trainability *on that tiny probe*.

Failure of 1A would block all downstream confirmatory interpretation. **1A passed.** Therefore 1B–1D maps may be interpreted as proposal evidence for the discovery atlas (subject to

the limitations in Chapter 12). The 1C freeze used a documented fallback after the strict intersection was empty. Experiment 1D is complete. Experiment 2, the 32M language-model study, has not been run.

Chapter 4

Parameterization and Terminology Glossary

This chapter is the dictionary. Later chapters assume it.

4.1 Coefficient tuple $(s, \alpha, \beta, \gamma)$

The family is

$$W_q = sA, \quad W_k = s(\alpha A + \beta B + \gamma C), \quad \alpha^2 + \beta^2 + \gamma^2 = 1.$$

A and C are Gaussian matrices with entry standard deviation $1/\sqrt{d}$. B is a random matrix whose row space is orthogonal to the row space of A , so $AB^\top = 0$ exactly. The four reported numbers are constrained: they are not a free 4-D box.

Each admissible initialization is encoded by:

- $s > 0$: **dimensionless fan-in multiplier**. Entries of A, C are drawn with std. $1/\sqrt{d}$, then $W_q = sA$. Thus the entry std. is $\sigma = s/\sqrt{d}$, and $s = 1$ is the architecture baseline (the decoder uses std. $= 1/\sqrt{d}$). s is *not* an entry standard deviation and is *not* a size-invariant softmax temperature. Think of s as “how many times louder than the usual fan-in Gaussian.”
- $\alpha \in [-1, 1]$: **entrywise correlation** between flattened query and key coordinates within a head (target marginal correlation of Q_{ab} and K_{ab}). Positive α tends to make a token more similar to itself in query-key space, which raises the diagonal logits. Negative α does the opposite.
- β, γ : split between a **row-space orthogonal** component B and an **independent Gaussian** component C in the construction $W_q W_k^\top = s^2(AB^\top + C)$ with constraints ensuring valid covariance. After the shared αA piece is chosen, the leftover energy must go somewhere. β puts it in a direction A cannot see. γ puts it in a fresh independent Gaussian.
- $\phi \in [0, \pi/2]$: angle parameterizing (β, γ) on the admissible circle at fixed α ,

$$\beta = \sqrt{1 - \alpha^2} \cos \phi, \quad \gamma = \sqrt{1 - \alpha^2} \sin \phi.$$

At $\alpha = \pm 1$, ϕ is redundant (all angles collapse to the same point); the grid retains one endpoint representative per sign.

Four named corners, used later as controls:

Name	(α, β, γ)	Everyday reading
Independent	$(0, 0, 1)$	Query and key unrelated
Orthogonal	$(0, 1, 0)$	Leftover energy in A 's blind directions
Tied	$(1, 0, 0)$	Key is a scaled copy of query
Anti-tied	$(-1, 0, 0)$	Key is a scaled negative copy of query

4.1.1 The two scales that people mix up

Proposition 1 gives

$$\mathbb{E}[\ell_{ij} \mid x] = \alpha s^2 \sqrt{m} c_{ij}$$

and a variance of the form $\text{Var}(\ell_{ij} \mid x) = s^4(\dots)$ with a d -dependent orthogonal term g_{ij} .

There are therefore *two* different “sizes” in the construction:

1. **Weight-entry size:** $\sigma = s/\sqrt{d}$. This is what an optimizer sees as the typical magnitude of W_q and W_k entries.
2. **Logit operating point:** the typical attention score scales like $s^2\sqrt{m}$ times a token similarity c_{ij} . This is what softmax sees.

Copying a numerical s to a new (d, m) therefore changes both $\sigma = s/\sqrt{d}$ and the logit operating point $s^2\sqrt{m}$. Atlas labels attached to (s, α, ϕ) at $(d, m, n) = (64, 16, 64)$ do not transfer to 1D's $(128, 32, 128)$ or to a 32M LM. A baseline with entry std. σ_0 has equivalent scale $s_0 = \sigma_0\sqrt{d}$.

4.1.2 s is already a $1/\sqrt{d}$ multiplier, and that is not why it fails to transfer

A natural misreading is: “ s does not transfer because it is an absolute variance, rather than a multiple of $1/\sqrt{d}$.” That is backwards from the constructor.

In code, A is drawn as a standard Gaussian divided by $\sqrt{d}$, then multiplied by s :

$$A_{ab} \sim \mathcal{N}\left(0, \frac{1}{d}\right), \quad W_q = sA, \quad \sigma = \frac{s}{\sqrt{d}}.$$

So s is already the fan-in multiple. It is *not* the entry standard deviation, and it is *not* the entry variance. $s = 1$ means “use the usual $1/\sqrt{d}$ Gaussian”; $s = 2$ means “twice that loud.” Entry variance is s^2/d , which still carries the $1/d$.

The transfer failure is a different fact: one number is being asked to preserve two sizes that scale differently when (d, m) change.

Convention	What stays fixed if you copy the number	What still moves
Copy s (what 1D did)	fan-in multiple of $1/\sqrt{d}$	entries quieter as $1/\sqrt{d}$; logits hotter as $\sqrt{m}$
Copy $\sigma = s/\sqrt{d}$ (absolute entry std.)	weight-entry noise	logits still grow as $\sigma^2 d \sqrt{m}$
Copy $s^2 \sqrt{m}$ (logit pref-actor)	softmax temperature at fixed α	entry noise $\sigma = s/\sqrt{d}$ changes

Making s an absolute variance would not make atlas labels transferable. If entries had a width-independent std. σ with no $1/\sqrt{d}$, the mean logit would still move as $\alpha \sigma^2 d \sqrt{m} c_{ij}$. One scalar cannot keep both the optimizer’s weight noise and softmax’s operating point fixed unless (d, m) stay put.

The coordinate that would preserve softmax temperature is $s^2 \sqrt{m}$ (equivalently $\sigma^2 d \sqrt{m}$), not s and not σ . That is not the coordinate the atlas stored.

In plain language. If you remember only one warning from this chapter, remember this: s is already a multiple of $1/\sqrt{d}$, not an absolute variance. $s = 2$ at width 64 is still not the same initializer as $s = 2$ at width 128, because the weights get quieter as $1/\sqrt{d}$ while the typical attention scores get hotter as $\sqrt{m}$.

4.2 Token geometry notation

For tokens x_i, x_j with Gram entries $v_i = \|x_i\|_2^2/d$, $v_j = \|x_j\|_2^2/d$, $c_{ij} = x_i^\top x_j/d$:

- v_i is a normalized squared length of token i . After LayerNorm-like processing it is often near 1.
- c_{ij} is a normalized inner product, a measure of how similar two tokens are.
- **Predicted logit mean/variance:** closed-form $\mathbb{E}[\ell_{ij}]$ and $\text{Var}(\ell_{ij})$ from Lemma 1 / Proposition 1 (verified in 1A).
- $\hat{\tau}$: row-centered logit scale (empirical std. dev. of active logits in a row). Softmax depends only on differences, so the row mean of logits does not matter; $\hat{\tau}$ measures how spread out the scores are after removing that mean.
- $\hat{\Delta}$: **diagonal gap**—mean diagonal logit minus mean off-diagonal logit on the active set. A large positive gap is the raw material of self-locking.

4.3 Attention-distribution metrics (1B)

For each attention row P_i over active keys $k_i > 1$:

Normalized entropy H_{norm}

$H(P_i)/\log k_i$. Equals 1 for uniform attention; near 0 for nearly one-hot rows. This is the main “how spread out is attention?” number. Dividing by $\log k_i$ makes the metric comparable across causal rows, which have different numbers of allowed keys.

Distance from uniform d_{unif}

RMS $\|P_{i\cdot} - u_i\|_2$ where u_i is uniform on the active support. Small means “close to equal weights.”

Maximum mass $\max_j P_{ij}$

Largest probability in the row (peakedness). If this is 0.9, one key owns 90% of the attention.

Self-mass P_{ii}

Diagonal attention probability (self-locking diagnostic).

Attention effective rank

$\text{erank}(P)/n$ where $\text{erank}(P) = \exp(-\sum_k \lambda_k \log \lambda_k)$ on singular values of the probability matrix. This is a matrix-level diversity measure, not the same thing as row entropy. Uniform unmasked attention can have high row entropy and still be rank-one as a matrix.

Jacobian operator norm $\|J\|_2$

Spectral norm of softmax Jacobian $J(P_{i\cdot}) = \text{diag}(P_{i\cdot}) - P_{i\cdot}P_{i\cdot}^\top$ (local sensitivity / gradient health). If this is tiny, a small change in logits barely changes attention, which can starve learning.

Update-to-weight ratio

$\|\nabla W\| \cdot \eta / \|W\|$ from a single backward step on a synthetic one-batch MSE probe (not full LM training). A diagnostic of “did anything move?” after one step.

Kernel metrics

Singular values of $M = W_q^\top W_k$ (operator norm, effective rank, spectral concentration). These describe the query-key product itself, before softmax.

4.4 Preregistered descriptive labels

Labels are assigned from **95% bootstrap confidence intervals** (2,000 seed-resampling draws per stratum in this run) with precedence

self-locked $\succ$ concentrated $\succ$ diffuse $\succ$ gradient-starved
 $\succ$ screened candidate $\succ$ boundary/uncertain.

Precedence means: if a cell satisfies more than one rule, the earlier name wins. Self-locked is diagnosed before concentrated, because a token attending to itself is a special kind of peak, not just “peaky attention” in general.

The bootstrap, in ordinary words: each cell was measured with 64 random seeds. We repeatedly redraw those 64 seeds with replacement, recompute the median metric, and look at the 2.5% and 97.5% percentiles of that list. If even the pessimistic end of that interval still clears a threshold, we treat the label as confirmed. If the interval sits on both sides of the threshold, we call the cell boundary / uncertain rather than forcing a name.

Diffuse

Lower CI bound on median $H_{\text{norm}} \geq 0.95$ and upper CI bound on $d_{\text{unif}} \leq 0.10$. Translation: we are confident attention is nearly uniform.

Concentrated

Upper CI bound on median $H_{\text{norm}} \leq 0.50$, *or* lower CI bound on median $\max_j P_{ij} \geq 0.80$. Translation: we are confident attention is peaky.

Self-locked Lower CI bound on median self-mass ≥ 0.80 . Translation: we are confident the token is mostly attending to itself.

Gradient-starved

Upper CI bound on Jacobian or update-to-weight below pilot floors (see Section 6.5). Translation: the one-step probe looks quieter than a baseline we measured independently.

Screened candidate

No pathology above confirmed; H_{norm} between 0.60 and 0.95; attention effective rank above pilot floor. Translation: not uniform mush, not a spike, not self-locked, and the attention matrix is not too collapsed, according to these screens.

Boundary / uncertain

Intervals cross thresholds (genuine measurement uncertainty).

Important: “Screened candidate” means “passes forward pathology screens,” not “will train well.” Trainability is Experiment 1D; the 1D gate is mixed (Chapter 9).

Chapter 5

Experiment 1A: Mathematical and Implementation Gate

In plain language. 1A is quality control. It asks whether the code implements the formulas we wrote on paper. It is not trying to discover a new attention phenomenon. If 1A had failed, we would have stopped.

5.1 Purpose

1A is a **blocking quality-control stage**. It does not discover scientific novelty; it verifies that the codebase correctly implements:

- exact logit mean and variance formulas (Lemma 1 / Proposition 1);
- energy identities and entrywise variance targets (so changing α does not secretly change how large the weights are);
- gradient/backprop identities (Proposition 4);
- common-bias cross-head statistic (Proposition 5);
- model integration (custom initializer, checkpoint survival, forward finiteness).

Why each of those matters:

- If the mean/variance formulas are wrong, later talk about “the theory predicted this operating point” is empty.
- If energies do not match, a comparison between $\alpha = 0$ and $\alpha = 1$ is confounded by scale.
- If gradients are wrong, later backward probes are wrong.
- If the initializer does not actually change W_q and W_k , or if checkpoints scramble the tensors, later training jobs are not testing the family we think they are testing.

5.2 Experimental design

5.2.1 Configuration grid

We vary the things the formulas say should matter, and we include both easy and hard token-pair correlations.

- Widths $d \in \{32, 64, 128\}$.
- Head ratios $m/d \in \{1/8, 1/4, 1/2\}$.
- Token-pair correlations $c_{ij} \in \{-0.5, 0, 0.5, 1.0\}$ (fixed synthetic pairs).
- 24 coefficient points per geometry: four canonical corners plus 20 interior (α, ϕ) points.

Total moment configurations: $3 \times 3 \times 4 \times 24 = 864$.

5.2.2 Monte Carlo protocol

“Monte Carlo” here means: draw many independent copies of (W_q, W_k) , compute the attention score for a fixed token pair, and compare the sample mean and sample variance with the closed-form prediction.

For each configuration:

1. Draw W_q, W_k from the admissible construction at the specified $(s, \alpha, \beta, \gamma)$.
2. Compute empirical mean and variance of $\ell = (W_q x)^\top (W_k y) / \sqrt{m}$ over increasing draws (minimum 20,000, up to 200,000).
3. Stop when $\text{SE}(\hat{\mu}) \leq 0.01 \sigma_{\text{pred}}$ and $\text{SE}(\widehat{\text{Var}}) \leq 0.01 \text{Var}_{\text{pred}}$. That is: keep sampling until the Monte Carlo noise is small compared with the predicted scale, so a mismatch cannot be excused as “we did not sample enough.”
4. Record energies, orthogonality $\|AB^\top\|_F / (\|A\|_F \|B\|_F)$, entry variances, empirical entry correlation vs. α .

5.2.3 Hypothesis tests

For each configuration we form two-sided normal approximate p -values for mean and variance. Across all 864 configurations (1,728 tests), **Holm–Bonferroni** correction controls family-wise error at $\alpha = 0.01$.

The multiple-testing issue, in ordinary words: if you run 1,728 tests at “5% significance,” you expect dozens of false alarms even if every formula is true. Holm–Bonferroni raises the bar so that the chance of *any* false rejection, across the whole family, stays at most 1%.

A configuration **passes** only if:

- neither corrected mean nor variance test is rejected;
- precision targets met;

- relative orthogonality $< 10^{-6}$;
- component energies within 1% of $s^2 m$ (and B, C within 1% of m);
- entry variances within 2% of s^2/d ;
- $|\hat{\rho}_{\text{entry}} - \alpha| \leq 0.02$.

5.2.4 Auxiliary checks (checks.json)

- **Gradient identity:** 20 random small configurations; analytic vs. finite-difference relative error $< 10^{-5}$.
- **Common-bias ratio:** empirical vs. predicted cross-kernel correlation ratio; error < 0.02 .
- **Baseline initializer match:** KS test comparing custom independent point to i.i.d. Gaussian baseline ($p > 0.01$, variance ratio in $[0.9, 1.1]$). This checks that our “independent corner” really is the ordinary Gaussian people already use.
- **Integration:** initializer changes W_q, W_k ; checkpoint reload is bitwise-equal; first forward pass finite.

5.3 Results

5.3.1 Gate decision

gate_summary.json: passed = true

Moment configurations tested: 864 Moment failures: 0 Gradient failures: 0

5.3.2 Moment accuracy

- All 864 configurations achieved `precision_converged = true`.
- Holm-rejected mean tests: **0 / 864**. Holm-rejected variance tests: **0 / 864**.
- Maximum relative orthogonality across configurations: $\lesssim 10^{-16}$ (numerical zero). The orthogonal construction is not “approximately orthogonal”; in floating point it is orthogonal.
- Uncorrected p -values span $(3 \times 10^{-4}, 1)$, consistent with valid null sampling rather than systematic bias. If the formulas were slightly but consistently wrong, p -values would pile up near zero. They do not.

Figure 5.1 is the actual moment-validation plot: empirical vs. predicted with a $y = x$ line, plus residuals. Calibration regression on the 864 configs is

$$\hat{\mu} = -3.4 \times 10^{-5} + 0.9999 \mu_{\text{pred}}, \quad R^2 = 0.99999, \quad \text{RMSE} = 0.0058,$$

$$\widehat{\text{Var}} = 0.0024 + 0.9976 \text{Var}_{\text{pred}}, \quad R^2 = 0.9975, \quad \text{RMSE} = 0.0106.$$

Median relative variance error is 0.62% (95th percentile 1.8%). This—not the 1B sequence scatter—is the evidence that Lemma 1 is implemented correctly.

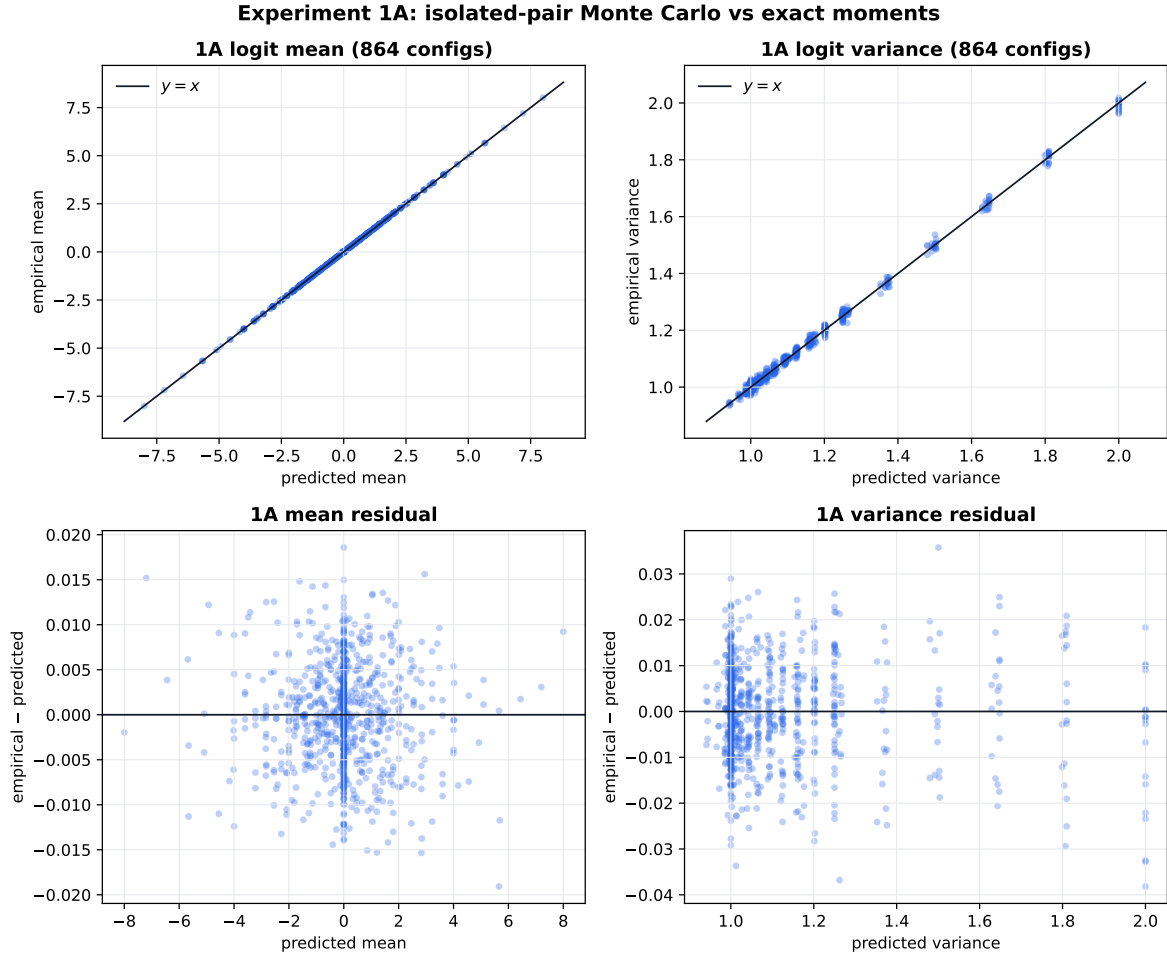


Figure 5.1: Experiment 1A: isolated-pair Monte Carlo moments versus exact theory. Each point is one of the 864 configurations. Points sit on the diagonal, and residuals are centered at zero with magnitude about 10^{-2} , matching the preregistered precision gate.

5.3.3 Auxiliary check highlights

- Common-bias absolute error: 1.76×10^{-4} (threshold 0.02).
- Baseline KS $p = 0.574$, variance ratio 1.009. The independent corner matches ordinary i.i.d. Gaussians.
- All 20 gradient configurations: identity error 0; finite-difference relative errors $\sim 10^{-10}$.
- Integration check: initializer changed tensors, checkpoint bitwise equal, forward finite.

5.3.4 Interpretation of 1A

The implementation faithfully realizes the finite-width theory within Monte Carlo tolerance. **Downstream atlas interpretation is unblocked.** Any future failure in 1B-Exp2 cannot be attributed to a Lemma 1 coding error without revisiting this gate.

What 1A does *not* say: it does not say that the formulas predict softmax entropy, trainability, or language-model quality. It says the first and second moments of isolated logits, plus a short list of identities, are implemented correctly.

Chapter 6

Experiment 1B: Discovery Atlas

In plain language. 1B is the map-making stage. We no longer look at one token pair in isolation. We run a one-block attention head on whole sequences, measure how attention looks, and name the regions.

6.1 Purpose

1B empirically maps **initialization-time attention behavior** across the admissible coefficient domain. Unlike 1A’s isolated token pairs, 1B runs a full forward pass through a one-block attention head on **sequences** of length $n = 64$ with value projection and (for the backward probe) a single optimization step.

The scientific product is a labeled atlas: for each allowed (s, α, ϕ) , under several input and mask conditions, we record whether attention is diffuse, concentrated, self-locked, screened, starved, or still too uncertain to name. That atlas is a measurement. It is not yet a training result.

6.2 Reference geometry and grid

Parameter	Value
Token dimension d	64
Head dimension m	16
Sequence length n	64
$\log_2 s$ grid	17 values in $[-2, 2]$
α grid	13 values in $[-1, 1]$
ϕ grid	9 values in $[0, \pi/2]$
Unique cells	1,717
Seeds per cell	64 paired seeds

Why this geometry: it is large enough that finite-width effects are visible, small enough that a dense three-dimensional grid with 64 seeds is affordable. s is gridded in log space because softmax cares about multiplicative scale. The cell count is 1,717 rather than $17 \times 13 \times 9 = 1,989$ because ϕ is redundant at $\alpha = \pm 1$ and those duplicates are not stored as separate scientific

points.

6.3 Measurement strata

Each cell is evaluated under **eight** input/mask strata (per ϕ slice, aggregated in analysis):

1. Synthetic inputs with token correlation $r = 0, 0.25, 0.5$ (LayerNorm applied). These are controlled clouds of vectors with a chosen typical similarity.
2. **Real activations**: post-LayerNorm tensors from an independently initialized baseline decoder (distribution shift test). This asks whether the map is an artifact of Gaussian toys.
3. **Causal** (lower-triangular) vs. **unmasked** (full) attention. Causal attention is what language models actually use; unmasked attention is a cleaner diagnostic because every row has the same number of keys.

Raw records per full run: $1,717 \times 64 \times 8 = 879,104$ rows (matches `analysis_summary.json`).

6.4 Analysis pipeline

1. Load all `atlas_shard*.jsonl` shards (~ 1.3 GB JSONL).
2. Compute pilot floors from independent baseline $(s, \alpha, \gamma) = (1, 0, 1)$ (Section 6.5).
3. Group rows by $(\text{cell_id}, \text{input_kind}, r, \text{mask}, d, m, n)$.
4. For each group and each of seven metrics, compute median and **cluster bootstrap 95% CI** with 2,000 draws resampling the 64 seeds.
5. Assign label + ± 0.05 sensitivity labels.
6. Export `atlas_summary.csv` (13,736 rows) and 72 stratum phase maps. The analyze-script residual scatter is *not* a 1A replication (Section 7.4).

The 13,736 summaries are 1,717 cells times 8 strata. Labels live at the stratum level because causal vs. unmasked, or synthetic vs. real, can change the picture.

6.5 Pilot thresholds

Derived from the independent baseline at $s = 1$ (5th percentile across baseline records):

Floor	Value
Jacobian operator norm	0.0556
Update-to-weight (min of W_q, W_k)	9.75×10^{-6}
Attention effective rank	0.275

These floors are not universal physical constants. They are “quieter than the ordinary independent initializer, as measured in this probe.” Gradient-starved labels are rare when floors are low; only **one** stratum in the entire atlas was labeled gradient-starved. That rarity is partly a fact about the family, and partly a fact about how low the floors sit.

Chapter 7

Experiment 1B Results

7.1 Global label histogram

Pooling all 13,736 stratified summaries:

Label	Count	Percent	Qualitative reading
Diffuse	5,362	39.0%	Near-uniform attention rows
Concentrated	3,793	27.6%	Low entropy or high peak mass
Screened candidate	2,867	20.9%	Mid-entropy, passes pathology screens
Boundary / uncertain	1,241	9.0%	Thresholds not resolved at 95%
Self-locked	472	3.4%	High diagonal self-attention
Gradient-starved	1	< 0.01%	Jacobian/update below pilot floor

The map is not a thin curve with everything else unlabeled. About two-fifths of measured strata are diffuse, about one-quarter are concentrated, and about one-fifth pass the screened-candidate screens. Self-locking exists but is uncommon. Gradient starvation at initialization, under these floors, is essentially absent.

That is already a scientific statement, but a limited one: it is a statement about *initialization-time attention on a one-block probe*, not about language modeling.

7.2 Synthetic vs. real activations

Input kind	Diffuse	Concentrated	Screened	Boundary	Self-locked	Grad-starved
Synthetic	4,115	2,811	2,115	960	300	1
Real activation	1,247	982	752	281	172	0

Screened fraction: **24.3%** synthetic vs. **21.5%** real (similar order of magnitude; real activations slightly shift mass toward concentrated/self-locked regions). The qualitative continents of the map survive the distribution shift. The coastlines move a little.

7.3 Reference stratum deep dive

We highlight **synthetic, unmasked**, $r = 0.25$, $\phi = 0$ —the canonical “paper slice” with 221 coefficient points (one per admissible (s, α) at this ϕ). This is the slice one would put in a main-text figure: not too correlated, not causal, orthogonal leftover turned off.

Label	n	Mean H_{norm}	Mean max mass	Mean self-mass	Role
Diffuse	99	0.990	0.028	0.017	High-entropy plateau
Screened candidate	40	0.829	0.141	0.038	Candidate band
Concentrated	60	0.186	0.761	0.076	Peaked attention
Self-locked	8	0.022	0.980	0.971	Diagonal collapse
Boundary	14	0.613	0.301	0.044	Uncertain

Read the means as a sanity check on the names. Diffuse rows really are almost uniform (entropy 0.99, max mass 0.028). Concentrated rows really are peaky (max mass 0.76). Self-locked rows really are looking at themselves (self-mass 0.97). Screened rows sit in between, with entropy 0.83 and a modest peak of 0.14. The labels are not an arbitrary coloring of a smooth field; they track different operating regimes.

At $\phi = \pi/4$ on the same stratum ($n = 187$ cells), counts are similar (85 diffuse, 33 screened, 53 concentrated, 3 self-locked), indicating ϕ rotates mass between orthogonal/independent components without collapsing the map.

7.4 Logit moments inside the 1B forward pass

This section exists because it is easy to misread a diagnostic plot as a second proof of 1A. It is not.

1A tests pairwise moments with many i.i.d. draws of (W_q, W_k) . 1B stores a *different* diagnostic: for each seed it compares one realized $n \times n$ logit matrix to the closed-form gram formulae. These quantities are not interchangeable.

What is stored. Empirical `logit_mean` / `logit_variance` are the sample mean and sample variance of the realized masked logits $\{\ell_{ij}\}$. Predicted `predicted_logit_mean` is the average of the pairwise conditional means $\mathbb{E}[\ell_{ij} \mid x]$. Predicted `predicted_logit_variance` is the average of the pairwise *conditional* variances $\text{Var}(\ell_{ij} \mid x)$, **not** the across-pair variance of the map.

The distinction in ordinary words:

- 1A asks: for one fixed pair of tokens, if I redraw the weights many times, do the mean and variance of that one score match the formula?
- 1B’s stored variance asks: in one forward pass, how spread out are all the scores in the sentence?

Those are different spreads. By the law of total variance,

$$\text{Var}_{ij}(\ell_{ij}) = \mathbb{E}_{ij}[\text{Var}(\ell_{ij} \mid x)] + \text{Var}_{ij}(\mathbb{E}[\ell_{ij} \mid x]).$$

The first term is “each pair is noisy given the tokens.” The second term is “different pairs have different average scores, because some tokens are more similar than others.” The second term is large whenever αs^2 is large (token Gram heterogeneity). Plotting empirical across-pair variance minus mean pairwise conditional variance therefore *must* show structured bands at discrete s values. That is an estimand mismatch, not a 1A failure.

The retracted statistic. The previous draft quoted mean residuals -0.0048 ± 0.0484 and $+0.0003 \pm 0.0013$ as “excellent agreement.” Those numbers used only the first 20,000 raw rows (cell ids 0–39, the smallest-scale cells). The analyze-script figure itself was drawn on all 879,104 records. The two do not match, and the global mean residual can cancel opposite stratum biases (e.g. $r = 0$ vs. $r = 0.5$). This paragraph is left in the long-form report so that the paper does not accidentally revive the retracted claim.

Full-data calibration (this report). On all 879,104 single-draw records, OLS of empirical vs. predicted is:

Estimand	a	b	R^2	RMSE	Median rel. err.
Mean (raw records)	0.004	1.001	0.788	1.77	4.7%
Mean (seed-averaged, 13,736 strata)	0.004	1.001	0.996	0.21	0.64%
Variance, pairwise formula (raw)	0.14	1.026	0.920	20.5	17%
Variance, pairwise (seed-averaged)	0.13	1.027	0.949	16.1	16%
Variance, total-variance correction (raw)	−0.06	0.975	0.936	18.3	14%
Variance, total-variance (seed-averaged)	−0.07	0.975	0.965	13.3	8.0%

Mean prediction is calibrated ($b \approx 1$). Single-draw residuals are wide (RMSE = 1.77); averaging 64 seeds recovers $R^2 = 0.996$. That supports the mean formula at sequence scale after Monte Carlo averaging, not per-record precision. One noisy sentence is not a 1A test. Sixty-four sentences, averaged, start to look like one.

Variance is **not** “excellent agreement.” The pairwise residual mean on the full file is $+0.96 \pm 20.4$, not $+0.0003$. Discrete islands in the residual plot coincide with the 17 values of s . Stratified pairwise variance residuals (raw records):

Stratum	Mean residual	RMSE	OLS slope
Synthetic inputs	−0.55	20.4	0.98
Real activations	+5.46	20.7	1.18
Unmasked	+0.27	18.8	1.00
Causal	+1.64	22.0	1.05
$r = 0$	+5.37	20.5	1.18
$r = 0.25$	+0.96	15.3	1.03
$r = 0.5$	−7.97	24.3	0.75

Positive and negative stratum biases cancel in the pooled mean. That is why a single global residual can look deceptively small. Adding an off-diagonal Gram estimate of $\text{Var}_{ij}(\mathbb{E}[\ell_{ij} \mid x])$ improves R^2 and centers residuals by s , but heteroscedasticity remains (spread grows with s).

The 1A formula is still the right pairwise theory; the 1B stored variance is a different functional of the sequence map.

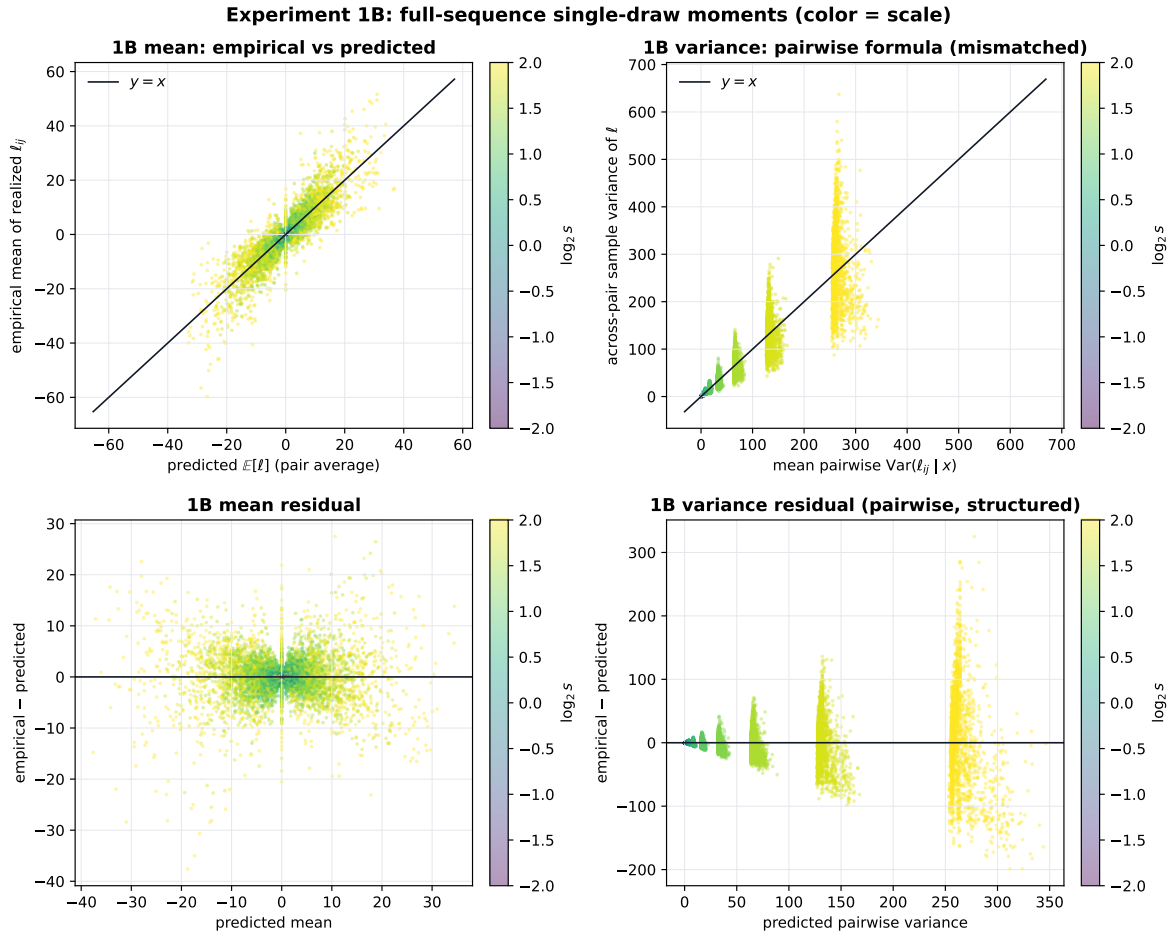


Figure 7.1: 1B single-draw diagnostics. Left: mean is approximately unbiased with heteroscedastic scatter. Right: empirical across-pair variance vs. mean pairwise conditional variance—mismatched estimands, discrete s -bands, large residuals. Color is $\log_2 s$.

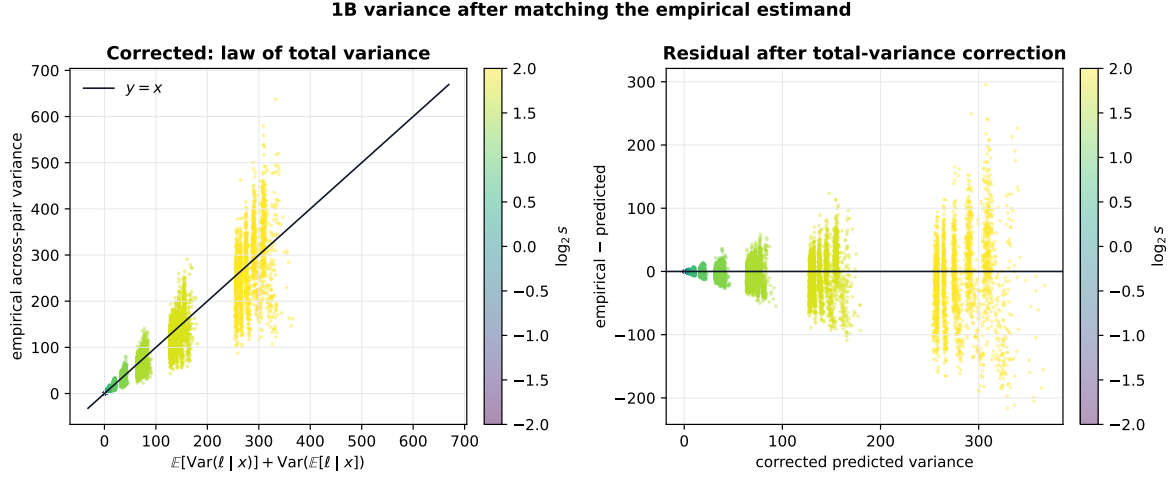


Figure 7.2: Same empirical variance after adding $\text{Var}_{ij}(\mathbb{E}[\ell_{ij} \mid x])$. Residuals are closer to centered by scale; remaining error is heteroscedastic single-draw noise, not 1A-style 10^{-2} Monte Carlo error.

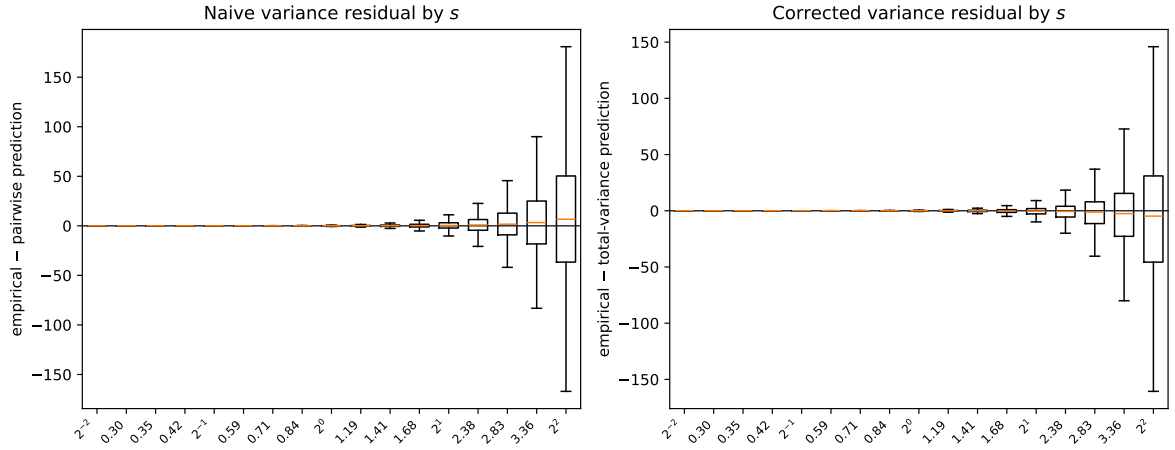


Figure 7.3: Variance residuals boxed by s . Naive pairwise residuals fan out and shift with scale; the total-variance correction recenters medians. Whiskers at $s = 4$ still span $\mathcal{O}(10^2)$.

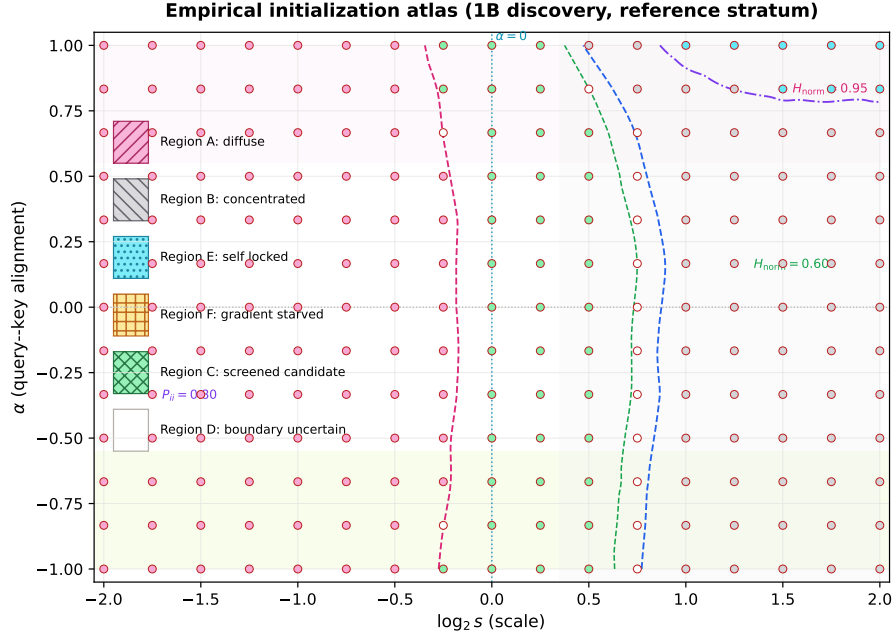
Conclusion for this section: 1A remains the confirmatory moment test. 1B means track the theory after seed averaging. 1B variances require matching the across-pair estimand and stratified reporting; they do not license a claim of excellent per-record sequence-scale variance agreement.

7.5 Phase-map visualization

Figure 7.4 follows the reference style used in information-theory phase diagrams: hatched region legend, dashed operational contours, and red-edged dots on the measured $(\log_2 s, \alpha)$ grid. Region C (screened candidate) is drawn as a **narrow** horizontal band to avoid overstating its spatial extent.

How to read the axes: moving right increases s (louder weights, hotter logits). Moving up increases α (more query-key tying, more diagonal bias). The interesting scientific object is

whether those two knobs produce continents with different attention names, or just a smooth gradient that we later discretized. The figures show continents, but the sensitivity analysis below shows that many cells live near the beaches.



Synthetic LayerNorm inputs, unmasked attention, $r = 0.25$, $\phi = 0$, $(d, m, n) = (64, 16, 64)$. Dashed contours: bootstrap median fields; red-edged dots: measured coefficient cells.

Figure 7.4: Region-style empirical atlas for the reference stratum. Dashed curves trace interpolated median H_{norm} and self-mass fields; colored dots are measured cells with preregistered labels.

Figure 7.5 shows the same data as a smooth surface (analogous to a 3D capacity/entropy landscape).

Normalized row-entropy surface over coefficient space

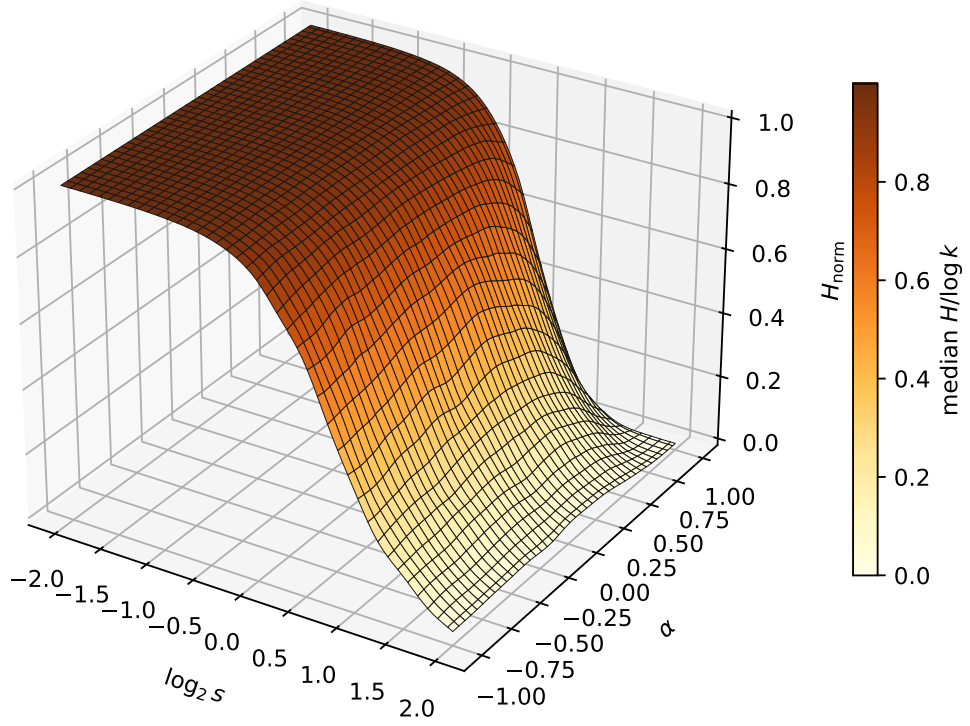


Figure 7.5: Interpolated normalized row-entropy surface over $(\log_2 s, \alpha)$. High plateau on the left is diffuse attention; the drop toward the right/top is concentration and self-locking.

Figure 7.6 gives the discrete nearest-cell label mosaic.

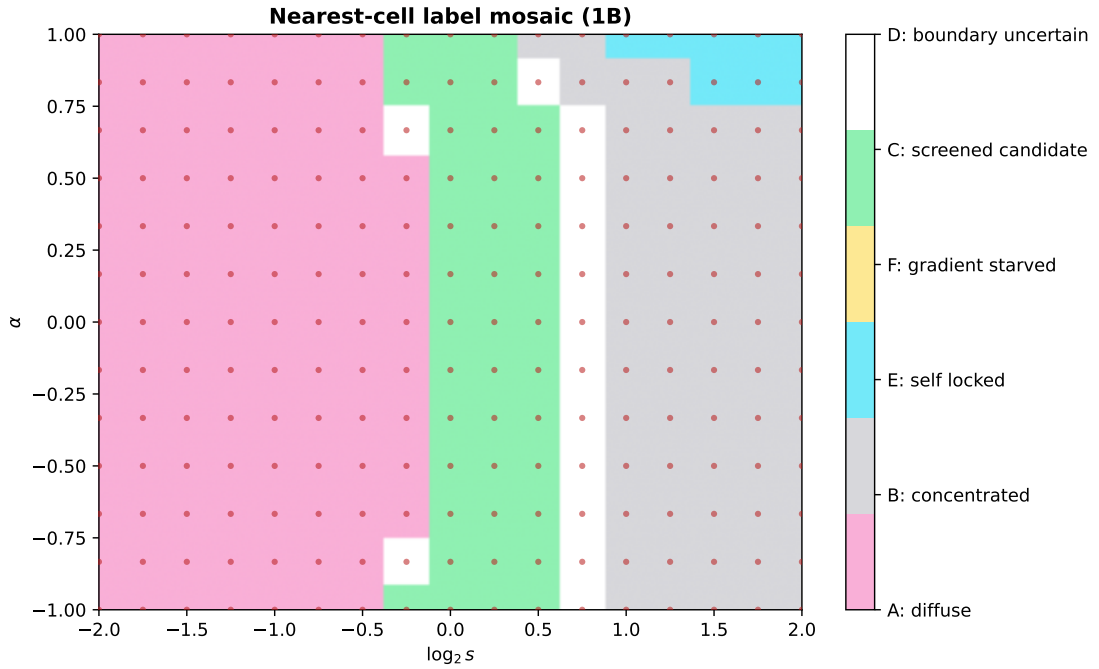


Figure 7.6: Nearest-measured-cell label assignment on the coefficient grid.

Legacy scatter plot (analysis script default):

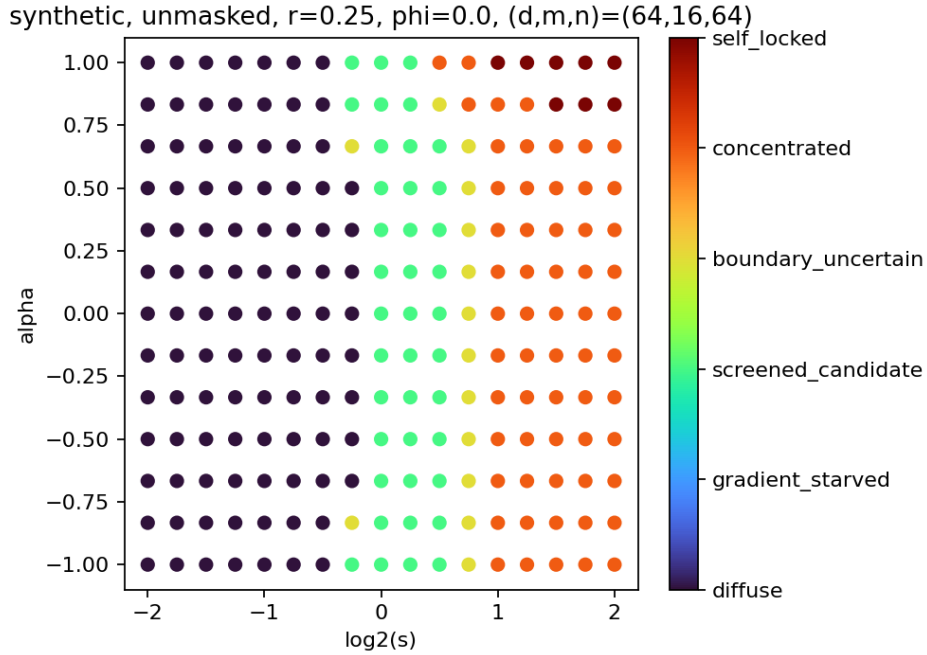


Figure 7.7: Original turbo-colormap scatter from `exp1b/run.py` analyze.

Additional 71 figures (varying ϕ , mask, r , and real activations) live under `code/exp1b/data/full/figures/`.

7.6 Label stability under threshold perturbation

Comparing primary labels to ± 0.05 probability/entropy shifts:

- 1,294 strata (9.4%) change label under -0.05 shift.
- 6,721 strata (48.9%) change under $+0.05$ shift.

Nearly half the atlas lies near label **boundaries** under generous threshold perturbation. This is expected: softmax metrics transition continuously. A 0.05 change in entropy or peak mass is not a different universe; it is a nearby contour. Stage 1C adaptive refinement was run to probe these boundaries (next chapter).

The paper should not describe the atlas as a set of sharp, robust countries. It is a useful map with wide coastal zones.

Chapter 8

Experiment 1C: Adaptive Refinement and Transfer

In plain language. 1C asks two practical questions. First: if we look harder at the uncertain parts of the 1B map, what do we find? Second: if we change width and sequence length, do the names stick? They mostly do not stick jointly. The eight-head file used later is a backup choice, not the original freeze rule.

8.1 Purpose

1C fits a probabilistic surrogate on 1B discovery labels, holds out 20% of coefficient cells, proposes 120 uncertain / theory-disagreement points, and remeasures them at

$$(d, m, n) \in \{(64, 16, 64), (128, 16, 128)\}$$

with 64 seeds and the same eight input/mask strata as 1B.

A “surrogate” here is a classifier trained to predict 1B labels from coefficients. It is a map interpolator, not a theory. We use it only to decide where to spend extra measurements.

The original freeze rule required a coefficient $(s, \alpha, \beta, \gamma)$ to be labeled `screened.candidate` on *every* replicated stratum. That is a high bar: eight strata at the reference geometry, and eight at the transfer geometry, all agreeing. After that intersection was empty, a documented fallback ladder froze the 8-head portfolio without looking at training outcomes.

Why freeze before training? If we pick “the heads that later trained well,” Experiment 1D becomes circular. The portfolio has to be chosen from initialization-time measurements only.

8.2 Compute that was merged

Kaggle produced two archives (`exp1c.tar`, `exp1cccc.tar`). Incomplete shards 0002 and 0003 from the first archive (5/30 tasks) were replaced by the completed copies. Effective sharding was 8 GPU streams covering all 240 tasks.

Item	Value
Adaptive cells	120
Geometries	2
Tasks	240 (all present)
Raw rows	122,880 (512 per task)
Stratified summaries	1,920
Bootstrap draws	2,000
Pilot floors	Jacobian 0.0241, update 5.94×10^{-6} , rank 0.175

The transfer geometry (128, 16, 128) doubles token width and sequence length but *does not* double head width. That is a different change from 1D, which uses (128, 32, 128). These three geometries should not be treated as one.

8.3 Surrogate held-out metrics

The classifier never trains on the held-out 20% of 1B cells. Reported metrics:

Metric	Value
Train / test rows	10,984 / 2,752
Balanced accuracy	0.987
Multiclass log-loss	0.0355
Expected calibration error	0.0153
Boundary error (top-quartile uncertainty)	0.0349

Balanced accuracy near 0.99 means: on held-out 1B cells, the interpolator almost always recovers the 1B name. Expected calibration error 0.015 means: when it says “80% chance screened,” it is about 80% right. The interpolator is well calibrated on 1B labels. It is **not** itself evidence that those labels are trainable, and it is not evidence that the names survive a geometry change. It only says the 1B map is smooth enough to interpolate.

8.4 Adaptive-cell labels

Label	All 1C	(64, 16, 64)	(128, 16, 128)
Diffuse	793	266	527
Boundary / uncertain	613	314	299
Screened candidate	514	380	134
Other pathologies	0	0	0

The transfer geometry is substantially more diffuse and less often screened. Majority-label agreement across the two geometries is 30/120 coefficient points. Only one quarter of the coefficient choices keep the same majority name after the transfer.

On the canonical 1B slice (synthetic, unmasked, $r = 0.25$, reference geometry) the 120 adaptive points are 94 diffuse, 25 boundary, and 1 screened: the sampler did what it was

asked to do—it spent budget on uncertain cells, not on the high-entropy plateau. That is why concentrated and self-locked counts are zero in 1C: those continents were not the sampling target.

8.5 How the portfolio was frozen

`freeze_portfolio` groups rows by $(s, \alpha, \beta, \gamma)$ and tries a documented ladder until at least 8 heads are available. Training outcomes are not used.

Freeze rule (120 coefficient points)	Count	Used?
All 16 replicated strata screened (preregistered)	0	no
Majority of all 16 strata screened	3	no (< 8)
Majority of 8 reference-geometry strata screened	32	yes
Any reference-geometry stratum screened	117	unused

Read the ladder slowly. The rule we wanted—screened in all 16 repeated settings—matched nobody. Even “screened in a majority of all 16” matched only three points, which is not enough to fill an eight-head portfolio. The rule that was actually used is weaker: among the eight strata at the *original* geometry (64, 16, 64), a point had to be screened more often than not. Thirty-two points passed; eight were taken by even coverage after sorting on (α, ϕ) .

The frozen file is `candidate_portfolio.json`. It records `frozen = true`, fallback rule `majority_reference_geometry`, `replication_strict = false`, and 32 eligible points. This is a **replication-strictness** fallback, not a new measurement. Experiment 1D was unblocked by this file plus a separate 1D selection that fills concentrated/self-locked cells from the 1B atlas (Chapter 9).

The paper must not say “we froze eight heads that replicated as screened across geometries.” That sentence is false. The true sentence is: “no such eight heads existed; we froze eight heads that were majority-screened on the reference geometry.”

8.6 Label sensitivity

Of 1,920 1C strata, 534 (27.8%) change label under a -0.05 probability/entropy shift and 1,404 (73.1%) change under $+0.05$. Adaptive cells sit even closer to decision boundaries than the 1B atlas, which is expected: we asked the sampler to go there.

Chapter 9

Experiment 1D: One-Block Trainability

In plain language. 1D is the first time anything is trained. The labels are locked before anyone looks at loss. We ask a narrow question: on a one-block decoder, do the cells we called screened improve faster than a pool of cells we called pathological? The answer depends on the task, and even the TinyStories “yes” is easy to over-read.

9.1 Purpose

1D tests whether initialization-time atlas labels predict **early trainability** of a one-block decoder. Coefficient points are frozen before any training loss is observed. The preregistered primary outcome is loss decrease per token over a fixed 2,000-step window. The atlas is called useful on a task only if screened candidates beat the pooled pathology groups (diffuse + concentrated + self-locked) with a paired 95% CI lying entirely above zero.

Why “loss decrease per token,” not final loss? Because runs can start at slightly different losses, and because the scientific object is early improvement, not a fully trained model. Why “per token”? Because the two tasks see different numbers of tokens, and a raw loss drop is not comparable until it is normalized.

Secondary outcomes: divergence, gradient clipping, final train loss, and whether the label predicts failure. This stage does *not* train a 32M language model (Experiment 2). It also does not evaluate held-out perplexity as a publication metric; it is a train-loss probe.

9.2 Frozen selection

Full selection required 15 diffuse, 15 concentrated, 15 self-locked, 15 screened, 20 boundary/uncertain points, plus four canonical corners at the median screened scale $s = 0.777$. Adaptive 1C cells have no concentrated or self-locked phenotypes (the sampler targeted uncertain cells), so those two strata were filled from the 1B discovery atlas. Labels were assigned before training; `selected_points.json` records `frozen_before_training: true`.

Stratum	Source atlas	n points	Median s
Diffuse	1C (majority on reference geometry)	15	0.802
Screened candidate	1C (majority on reference geometry)	15	0.777
Boundary / uncertain	1C (majority on reference geometry)	20	0.799
Concentrated	1B discovery fill	15	2.000
Self-locked	1B discovery fill	15	2.828
Canonical corners	matched to median screened s	4	0.777
Total		84	—

This table is the most important confound in 1D, and it should be in the paper. Screened, diffuse, and boundary points come from 1C and sit near $s \approx 0.8$. Concentrated and self-locked points come from 1B and sit at $s = 2.0$ and $s = 2.83$. The pathology *pool* therefore mixes “different attention shape” with “much larger s .” Figure 9.1 shows that 1C adaptive points form a narrow band near $\log_2 s \approx -0.35$, while 1B pathology fill occupies $s \geq 1$. Those numerical s values are **not** interchangeable across model sizes.

9.3 s does not transfer across (d, m)

The constructor is already fan-in scaled: A_{ab} has std. $1/\sqrt{d}$, then $W_q = sA$, so

$$\sigma = \frac{s}{\sqrt{d}}.$$

This is the same point as Section 4.1.2. s is *not* an absolute variance. The non-transferability is not “we forgot to divide by $\sqrt{d}$.” The $1/\sqrt{d}$ is already in A . The remaining problem is that softmax does not see σ ; it sees $s^2\sqrt{m}$.

The 1D decoder baseline uses exactly this σ at $s = 1$. Proposition 1 then gives a different size dependence for attention logits,

$$\mathbb{E}[\ell_{ij} \mid x] = \alpha s^2 \sqrt{m} c_{ij}.$$

Keeping the atlas number s fixed while changing width therefore does two different things at once: entries get quieter as $1/\sqrt{d}$, while the mean logit prefactor $s^2\sqrt{m}$ tracks head width. If instead one copied the absolute entry std. σ , the weights would match and the logits would still change, because

$$\mathbb{E}[\ell_{ij} \mid x] = \alpha \sigma^2 d \sqrt{m} c_{ij}.$$

1B labels were assigned at $(d, m, n) = (64, 16, 64)$. 1D trains at $(128, 32, 128)$ (d and m both doubled; 1C’s transfer geometry was $(128, 16, 128)$, a third pair). Median operating points:

Stratum	Median s	σ at 1B (64, 16)	σ as run in 1D	$s^2\sqrt{m}$ at 1B	$s^2\sqrt{m}$ in 1D
Screened / corners	0.777	0.097	0.069	2.41	3.41
Diffuse	0.802	0.100	0.071	2.57	3.64
Fan-in baseline	1.000	0.125	0.088	4.00	5.66
Concentrated	2.000	0.250	0.177	16.0	22.6
Self-locked	2.828	0.354	0.250	32.0	45.3

Transplanting $s = 2$ from 1B into 1D *lowers* entry std. ($0.250 \rightarrow 0.177$) and *raises* the logit-mean prefactor ($16 \rightarrow 22.6$). The 1B “concentrated” / “self-locked” labels therefore do not name the 1D initialization. Within 1D, every job shares $(128, 32)$, so ranking s is still a fair within-model scale comparison; it is not a transferable atlas coordinate for Experiment 2.

A useful translation of the table: in 1D, self-locked fill is initialized with entries about $3.6\times$ noisier than screened fill ($0.250/0.069$), and with a logit prefactor about $13\times$ larger ($45.3/3.41$). That is a different experiment from “same scale, different query-key correlation.”

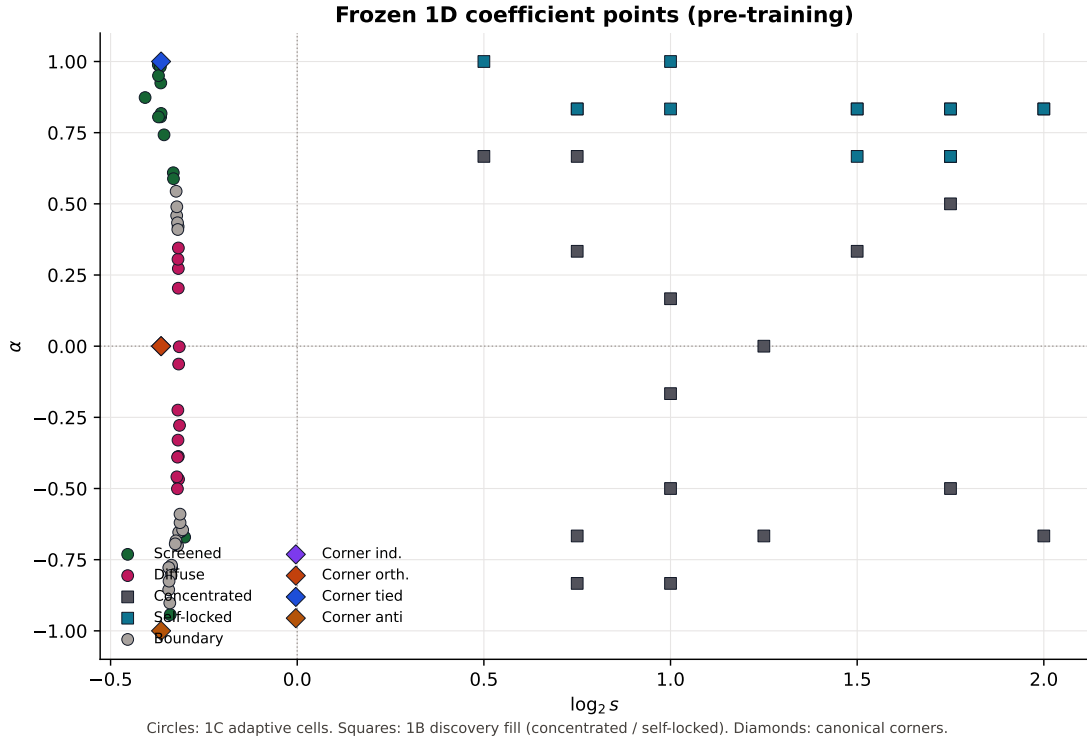


Figure 9.1: Frozen 1D coefficient points. Circles are 1C adaptive cells; squares are 1B concentrated/self-locked fill; diamonds are canonical corners. Independent and orthogonal corners share $\alpha = 0$ at the matched scale and overlap. The x -axis is the dimensionless s copied from the atlas, not $\sigma = s/\sqrt{d}$.

9.4 Training protocol

Each job is an independent (point, task, seed) run of the same one-block decoder:

Setting	Value
Architecture	1 layer, width 128, 4 heads, vocab 256, dropout 0
Context / batch	128 / 64
Steps / LR	2,000 / AdamW 3×10^{-4} , grad-clip 1.0
Paired seeds	20260829–20260836 (matched data order and non-QK params)
Tasks	associative recall; TinyStories byte-token LM
Tokens / job	128,000 (recall); 16,384,000 (TinyStories)
Query/key init	frozen $(s, \alpha, \beta, \gamma)$ applied to all 4 heads
Jobs	$84 \times 2 \times 8 = 1,344$

Two comments for later paper writing:

- The decoder is tiny on purpose. Residual streams, depth, and a large corpus are Experiment 2 problems.
- Seeds are paired: the same eight seeds, the same data order, and the same non-query/key parameters, across points. That is why the gate can subtract screened from pathology *within seed* rather than mixing unrelated training noise.

Training was packed across Kaggle 2-GPU kernels (8 notebook shards $\times$ 16 workers). After merge: 1,344/1,344 unique jobs, 0 divergences, 50 duplicate rows dropped from overlapping dumps. Analysis used 10,000 cluster-bootstrap draws on the 8 paired seed-level contrasts.

9.5 Primary gate

For each seed and task, the contrast is

$$\Delta_s = \overline{\text{loss decrease/token}}_{\text{screened},s} - \overline{\text{loss decrease/token}}_{\text{pathology},s}.$$

Pathology is the pool of diffuse, concentrated, and self-locked points. Boundary and corners are recorded but excluded from the gate.

In ordinary words: pick a seed. Average how much loss dropped per token among the 15 screened points. Subtract the same average among the 15 + 15 + 15 pathology points. Do that eight times, once per seed. The reported $\hat{\Delta}$ is the mean of those eight numbers. The CI is a bootstrap of those eight numbers. The gate passes only if even the pessimistic end of the CI is still positive.

Task	$\hat{\Delta}$	95% CI	Seeds	Gate	$\hat{\Delta} \times \text{tokens}$
Associative recall	-1.16×10^{-7}	$[-4.68, +2.47] \times 10^{-7}$	8	FAIL	-0.015 nats
TinyStories	$+2.06 \times 10^{-8}$	$[+1.86, +2.32] \times 10^{-8}$	8	PASS	$+0.338$ nats

The last column multiplies the per-token difference by tokens seen, so the reader can see a total-loss gap. TinyStories’ +0.338 nats is a small but consistent edge over 16.4 million tokens. Associative recall’s -0.015 nats is a rounding-error difference over 128 thousand tokens, and the CI includes both signs.

All eight TinyStories seed-level Δ_s are positive (range $[1.73, 2.81] \times 10^{-8}$). Associative-recall seed-level signs split 2–6. Figure 9.2 uses separate x -scales because the TinyStories interval is two orders of magnitude tighter than the recall interval.

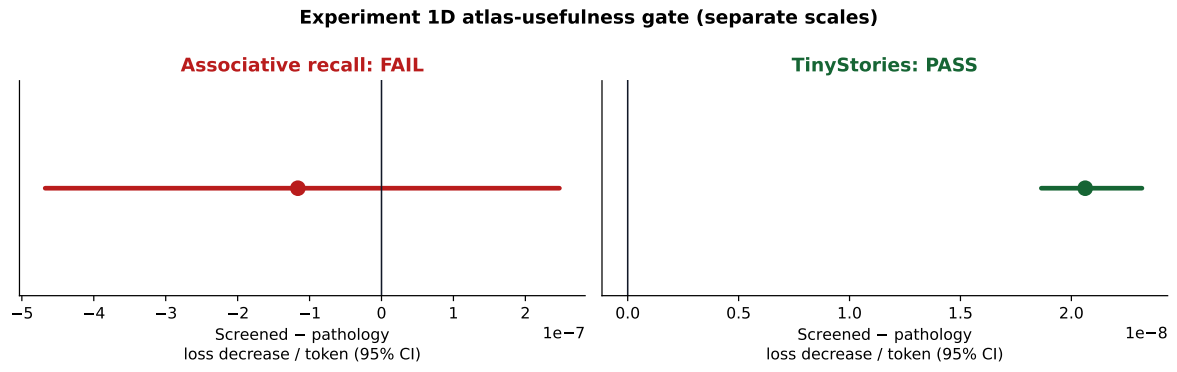


Figure 9.2: Preregistered atlas-usefulness gate. A task passes only when the paired CI for screened minus pathology loss decrease per token lies entirely above zero.

Chapter 10

Experiment 1D Results

10.1 Associative recall is not a discriminating probe

Every recall run clips gradients on 100% of steps and finishes at mean train loss 4.172 regardless of stratum (range of stratum means: 4.171–4.172). Mean total drop is 1.83–1.89 nats. Screened is not better than pathology; the CI includes zero and the point estimate is slightly negative (-0.015 nats over 128,000 tokens).

Associative recall, in this implementation, is a synthetic key-value memory task: the model sees pairs and must later retrieve the value given the key. A good probe would separate initializations. This one did not, but the reason matters. Gradient clipping frequency 1.0 means the optimizer wanted a larger step than the clip allowed, on every step, for every initialization. Final losses sit on top of each other. The task is clip-saturated.

The recall task therefore does **not** license a claim that atlas labels predict trainability. It also does not license the opposite claim: the probe saturates and is clip-bound, so stratum differences may be invisible rather than absent. For the paper: either retune the probe, or drop it as uninformative. Do not write “all initializations work equally well on recall.”

10.2 TinyStories: high-scale pathology trains worse

TinyStories is a dataset of short children’s stories. Here it is trained as a byte-level language model: predict the next byte. Unlike recall, this task does discriminate. Mean final train loss:

Stratum	Mean final loss	Mean total drop	Clip frequency
Screened candidate	1.360	4.678	0.061
Independent corner	1.361	4.678	0.087
Orthogonal corner	1.361	4.669	0.084
Tied corner	1.360	4.682	0.030
Boundary / uncertain	1.364	4.668	0.118
Diffuse	1.365	4.667	0.115
Anti-tied corner	1.366	4.662	0.157
Concentrated	1.619	4.410	0.013
Self-locked	2.095	3.944	0.013

There are two stories in this table, and they must not be fused.

Story A, the preregistered one. Screened minus pooled pathology is +0.338 nats over 16.384M tokens, with a CI that excludes zero. That is the preregistered pass. It is a true statement about the pooling rule we wrote down in advance.

Story B, the exploratory one. Screened versus diffuse (both $s \approx 0.79$, hence both $\sigma \approx 0.07$ at $d = 128$) differs by only 0.011 nats of total drop and 0.005 nats of final loss. Tied, independent, and orthogonal corners at the same s sit on top of screened and diffuse. The cells that lag are concentrated and self-locked, which are the 1B fill at much larger s . The gate is carried by 1B fill with larger dimensionless s , which at 1D is still a larger multiple of $1/\sqrt{d}$ ($\sigma = 0.177$ and 0.250 vs. 0.069) and a hotter $s^2\sqrt{m}$. Structure at matched fan-in multiplier is not the TinyStories story; **a larger multiple of $1/\sqrt{d}$, transplanted from a different (d, m)** is. Those 1B phenotype names should not be treated as 1D or Exp2 labels.

Clip frequencies add a small extra clue: high- s concentrated/self-locked runs clip *less* often (0.013) than the $s \approx 0.8$ band, yet they finish worse. The failure is not “they exploded and hit the clip.” They improved less.

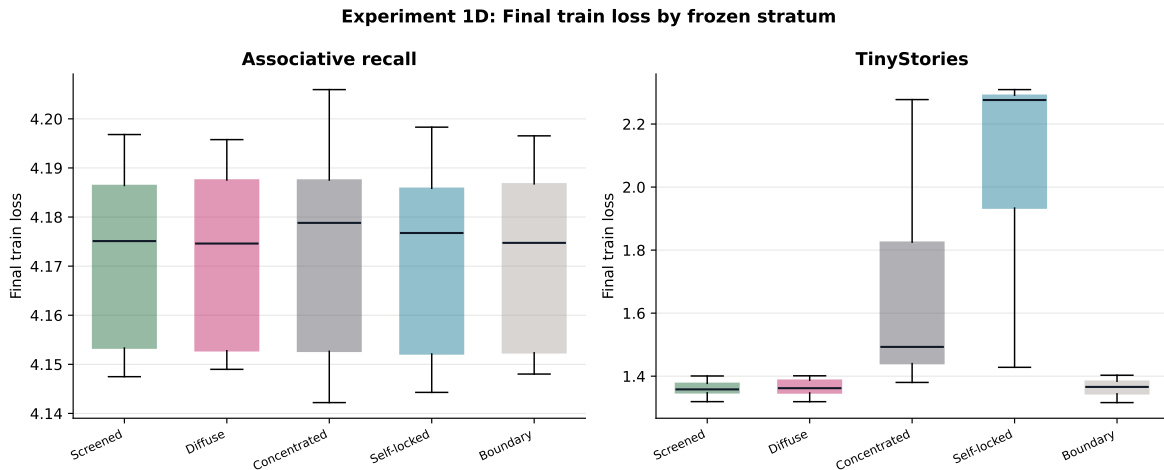


Figure 10.1: Final train loss by frozen stratum. Recall is flat. TinyStories separates high-scale concentrated/self-locked points from the $s \approx 0.8$ band.

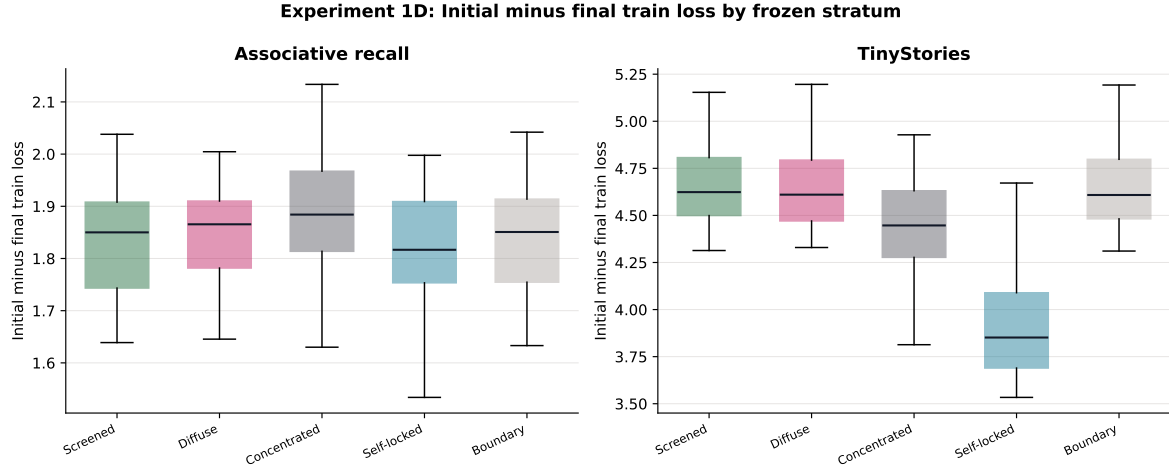


Figure 10.2: Initial minus final train loss. Larger is more early improvement. Self-locked TinyStories runs improve least.

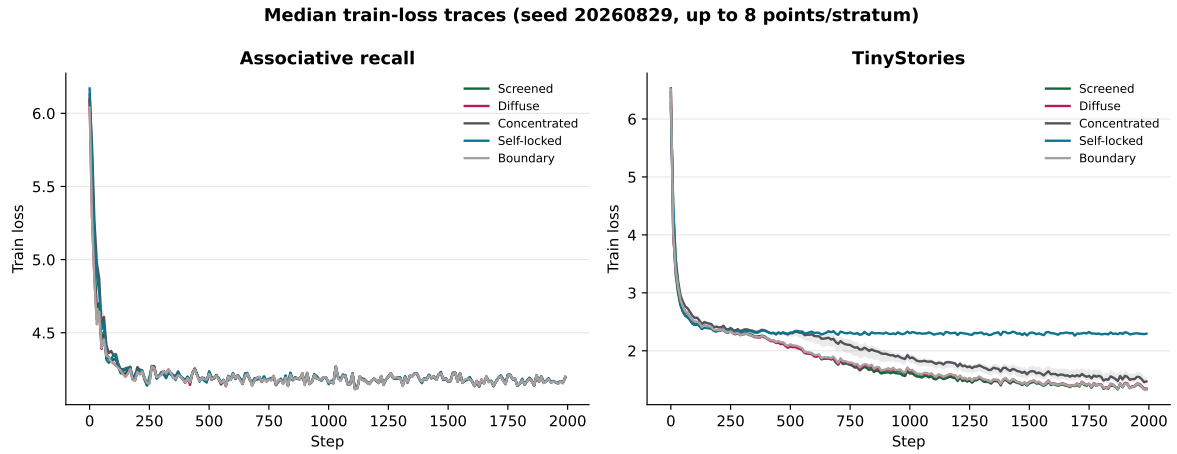


Figure 10.3: Median traces for seed 20260829 (up to eight points per stratum). Recall collapses to a common plateau by step ~ 100 . TinyStories self-locked plateaus near 2.3; concentrated lags; screened/diffuse/boundary continue descending together.

10.3 What 1D may and may not claim

- **May claim:** under the preregistered pooling rule, TinyStories screened points beat pooled pathology on early loss decrease; associative recall does not.
- **May claim:** one-block TinyStories training is slower to improve from the transplanted high- s 1B fill ($s = 2.0$ and 2.83 , i.e. $\sigma = 0.177$ and 0.250 at $d = 128$) than from the $s \approx 0.8$ 1C band ($\sigma \approx 0.07$).
- **Must not claim:** a numerical s from 1B is the same initializer at another width. s is already a $1/\sqrt{d}$ multiplier, not an absolute variance; the remaining mismatch is that logits scale as $s^2\sqrt{m}$ (or $\sigma^2 d\sqrt{m}$). 1B concentrated/self-locked labels do not transfer to 1D or Experiment 2.
- **Must not claim:** screened geometry is a better initializer than diffuse geometry at matched scale. That contrast is ≈ 0.005 nats of final TinyStories loss.

- **Must not claim:** the atlas is a validated 32M-LM treatment. Experiment 2 has not been run. The 1C 8-head portfolio is frozen under a fallback, not under the empty all-stratum intersection.
- **Must not claim:** associative recall “works equally well for all initializations.” The probe is clip-saturated.

No run diverged (0/1,344). Stability in the “did not explode” sense is real, and modest. It is not a quality result.

Chapter 11

Synthesis and Scientific Reading

This chapter is the one a later paper should compress, not replace.

11.1 What 1A established

The finite-width algebraic theory is not merely analytic—it is implemented correctly in floating point with rigorous Monte Carlo and gradient checks. This is a prerequisite most initialization papers skip. It licenses later sentences of the form “the code is computing the family we defined.” It does not license later sentences of the form “therefore these initializations train well.”

11.2 What 1B established

1. The admissible (s, α, ϕ) domain admits **large, structured regions** of distinct initialization phenotypes (diffuse vs. concentrated vs. screened). The map is not a single blob.
2. **Self-locking** is measurable but occupies a minority ($\sim 3.4\%$ of strata).
3. **Gradient starvation** at initialization is extremely rare under the pilot floors (1 stratum).
4. Pairwise mean identities carry over to 1B after seed averaging ($R^2 = 0.996$). Sequence-scale *variance* as stored in 1B is a different estimand and is only approximately calibrated after a total-variance correction (Section 7.4); it does not re-prove 1A.
5. Real activations preserve qualitative label proportions but redistribute mass toward concentration relative to synthetic Gaussians.
6. Many cells sit near label boundaries: a 0.05 threshold nudge relabels nearly half the atlas in one direction. The continents are real; the borders are thick.

11.3 What 1C established

1. A discovery surrogate can be fit and held out with balanced accuracy 0.987 and ECE 0.015. The 1B map is interpolable.
2. The 120 adaptive cells were fully remeasured at two geometries (122,880 rows).

3. Screened labels are **not** stable under mask, correlation, real activations, and width/length transfer jointly: the preregistered freeze intersection is empty (0/120).
4. Transfer to (128, 16, 128) shifts mass toward diffuse (527 vs. 266 at reference).
5. A documented fallback froze 8 heads from 32 majority-screened reference-geometry points (`replication_strict: false`).

11.4 What 1D established

1. 84 points were frozen before training and all 1,344 jobs completed with zero divergence.
2. Associative recall does not pass the atlas-usefulness gate (CI includes 0; every step clipped; all strata finish at 4.172).
3. TinyStories passes the preregistered pooled-pathology gate (+0.338 nats; CI excludes 0).
4. The TinyStories pass is a within-1D larger fan-in-multiplier deficit, not a screened-versus-diffuse geometry effect, and not a size-invariant s law. 1B numerical s is not transferable even though s is already a $1/\sqrt{d}$ multiple: $\sigma = s/\sqrt{d}$ and $\mathbb{E}[\ell] \propto s^2\sqrt{m}$. The failure is not “ s was an absolute variance.”

11.5 What is *not* yet established

- An 8-head `candidate_portfolio.json` under the original all-stratum intersection (still 0 points).
- Whether screened geometry beats diffuse geometry at matched scale (1D: ≈ 0.005 nats TinyStories final-loss gap).
- Whether a diverse portfolio improves a 32M LM (Experiment 2).
- Sign-symmetry unit test on 50 cells (`sign_symmetry_check.json` not present in the full 1B analyze artifact).

11.6 A candidate paper spine

If this memoir is later shortened, a defensible narrative spine is:

1. Joint query-key initialization is a real, energy-matched degree of freedom, with exact finite-width logit moments (1A).
2. That family produces an empirical initialization atlas with diffuse, concentrated, self-locked, and mid-entropy screened regions (1B).
3. Screened names do not jointly survive a strict multi-stratum plus transfer freeze (1C).
4. On a one-block decoder, early TinyStories trainability tracks transplanted scale more than screened-versus-diffuse structure; associative recall is an uninformative saturated probe (1D).

5. A 32M confirmatory study remains necessary, and must convert scale with $s_0 = \sigma_0\sqrt{d}$ rather than pasting 1B's numerical s .

Anything stronger than that spine is not supported by this report.

Chapter 12

Limitations and Provenance

- **Single-block probe:** no depth, no residual stream, no full LM in 1B. The atlas is an initialization snapshot of one attention block.
- **One backward step:** update ratios are diagnostic, not training trajectories.
- **Bootstrap draws:** 2,000 (manifest `exp1b_1788602938369_18144.json`); coarser than pre-registered 10,000 but sufficient for exploration.
- **CPU/RAM analyze:** ~ 1.3 GB JSONL loaded fully into memory.
- **Labels are descriptive,** not causal claims about generalization.
- **1B logit variance is not the 1A estimand:** `predicted_logit_variance` averages pairwise conditional variances; `logit_variance` is the across-pair sample variance of one realized map. Pooled mean residuals cancel opposing r and input-kind biases.
- **1C freeze used a fallback:** a point must be screened on all 16 replicated strata to be `replication_strict`. Zero of 120 points pass that rule. The frozen 8-head file uses majority screening on the reference geometry (64, 16, 64).
- **Numerical s is not transferable across sizes, and not because it is an absolute variance:** the constructor already uses $1/\sqrt{d}$ ($\sigma = s/\sqrt{d}$; entry variance s^2/d). 1B labels live at (64, 16, 64); 1D ran (128, 32, 128). Same s is a different entry std. and a different $s^2\sqrt{m}$ logit mean. Copying absolute σ instead would still move logits as $\sigma^2 d \sqrt{m}$. Experiment 2 must convert a baseline σ_0 via $s_0 = \sigma_0 \sqrt{d}$ and must not paste 1B s as a temperature. See Section 4.1.2.
- **1D pathology pool mixes scale with phenotype:** concentrated/self-locked 1D cells come from 1B at median $s = 2.0$ and 2.83; screened/diffuse come from 1C at $s \approx 0.8$. Even after $1/\sqrt{d}$, those groups are not scale-matched at 1D.
- **1D associative recall is clip-saturated:** gradient clipping frequency 1.0 on every recall job; stratum means of final loss differ by < 0.002 nats.
- **1D is a one-block, 2,000-step train loss study,** not held-out 32M LM quality.
- **Git LFS:** raw shards require `git lfs pull` or manual upload on cloud machines.

Artifact paths (all under the repo `code/` and `output/pdf/` trees):

- 1A full: `exp1a/data/full/`
- 1B full: `exp1b/data/full/`
- 1C full: `exp1c/data/full/` (includes the frozen portfolio JSON)
- 1D full: `exp1d/data/full/` (selection, analysis, training shards)
- 1D numeric dump: `output/pdf/exp1d_diagnostics.json`

Chapter 13

Recommended Next Steps

1. **Experiment 2:** 32M LM confirmatory study. Keep s as a fan-in multiplier ($s = 1$ means $\sigma = 1/\sqrt{d}$ at the Exp2 width; baseline equivalent is $s_0 = \sigma_0\sqrt{d}$). Do not recode s as an absolute variance to “fix transfer”: that would still leave logits scaling as $\sigma^2 d \sqrt{m}$. Do not copy 1B/1C numerical s as if it preserved softmax temperature when m changes. Prefer contrasts that match σ (equivalently s) at the Exp2 geometry: screened vs. diffuse vs. independent corner. Do not treat the 1D TinyStories pooled-pathology pass as a substitute, and do not treat 1B concentrated/self-locked labels as Exp2 treatments.
2. **Optional 1D follow-up:** concentrated/self-locked cells remeasured at $(d, m) = (128, 32)$ before training, or a pathology set with the same s (same $1/\sqrt{d}$) as the screened band. The current 1B fill cannot separate phenotype from a transplanted scale.
3. **Recall probe:** if associative recall is kept, retune learning rate / clip / pair count so the task is not 100% clip-bound.
4. **Paper condensation:** start from Chapter “Synthesis and Scientific Reading,” keep the mixed 1D result and the 1C fallback, and move grids, freeze ladders, and moment-estimand diagnostics to an appendix or supplement.

Appendix A

Holm–Bonferroni reminder

Given ordered p -values $p_{(1)} \leq \dots \leq p_{(m)}$, reject hypothesis i if $p_{(i)} \leq \alpha/(m - i + 1)$. We applied $m = 1728$, $\alpha = 0.01$ in 1A.

In ordinary words: sort the 1,728 p -values from smallest to largest. The smallest one must beat $0.01/1728$ to count as a rejection. The next must beat $0.01/1727$, and so on. This is stricter than uncorrected 0.01 testing, and it is why “0 of 864 mean tests rejected” is a stronger statement than it first appears.

Appendix B

Bootstrap procedure in 1B

For each stratum with seed values $\{x_s\}_{s=1}^{64}$:

1. Resample 64 seeds with replacement $B = 2000$ times.
2. Compute median metric per resample.
3. Report 2.5% and 97.5% quantiles as CI endpoints.

Seeds are paired across cells; heads/tokens nested within seed are not separate bootstrap units (cluster bootstrap at seed level).

Why cluster at seed level? Because tokens and heads from the same random draw are not independent replicates. Pretending they are would make confidence intervals look narrower than they are.

Appendix C

1D paired bootstrap

For each task, form eight seed-level contrasts Δ_s as the difference of mean screened and mean pathology loss-decrease-per-token among non-divergent jobs with that paired seed. Then:

1. Point estimate: mean of $\{\Delta_s\}_{s=1}^8$.
2. Resample the eight contrasts with replacement $B = 10,000$ times; 2.5% / 97.5% quantiles are the CI.
3. Gate: pass iff the lower endpoint is strictly positive.

Boundary and canonical-corner jobs are excluded from Δ_s . Implied total-loss gap is $\hat{\Delta}$ times mean tokens seen (128,000 recall; 16,384,000 TinyStories).

The eight seed-level numbers are the actual experimental replicates. Points inside a seed are averaged first so that a stratum with 15 points does not pretend to be 15 independent worlds.